INCH-POUND

MIL-C-55302/178A
26 DEC 1989
SUPERSEDING
MIL-C-55302/178
15 April 1987

MILITARY SPECIFICATION SHEET

CONNECTORS, PRINTED CIRCUIT SUBASSEMBLY AND ACCESSORIES: 3-ROW PIN ASSEMBLIES, RIGHT ANGLE, 75 THROUGH 405 CONTACT POSITIONS, FOR PRINTED WIRING BOARDS (.100 SPACING)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation MIL-C-55302.

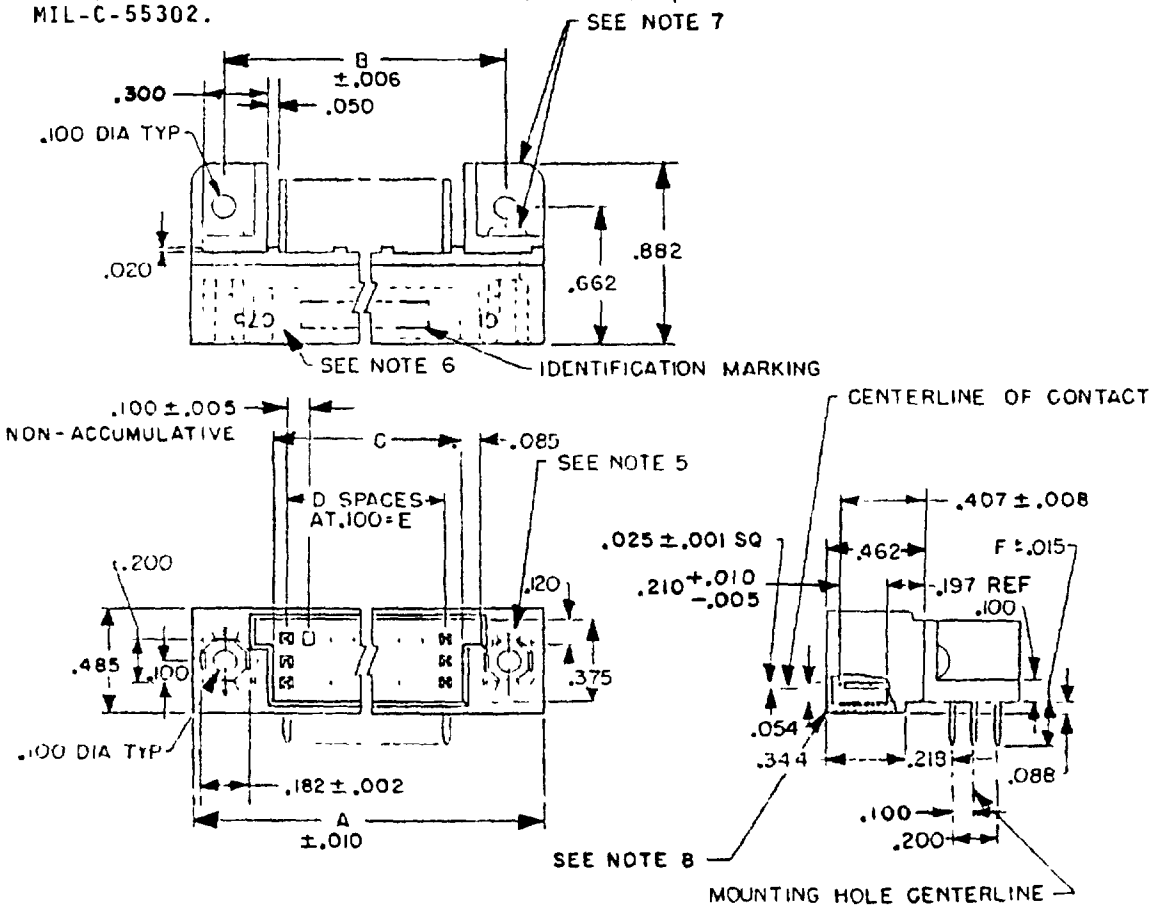
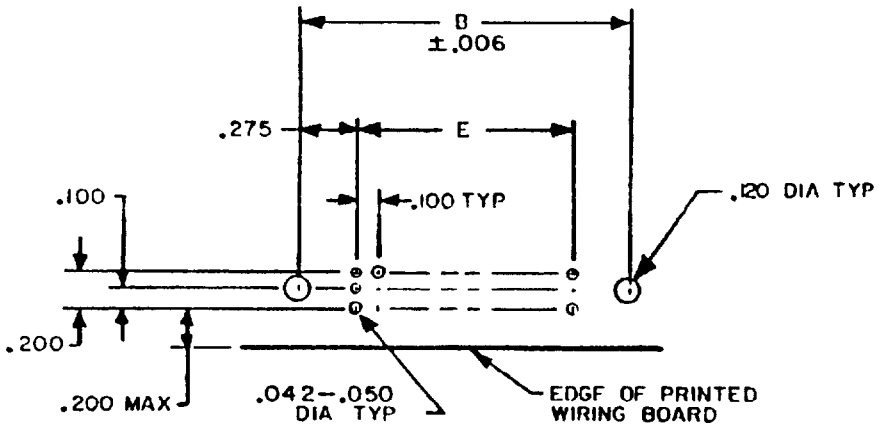


FIGURE 1 Connectors, pin assemblies, right angle .100 (2.54 mm) spacing, 75 through 225 contacts.

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Recommended circuit layout

F ±.015 (0.38 mm)		Type	Type of termination
Inches	mm		
.180	4.57	I	Solder post
.120	3.05	IV	

Inches	mm	Inches	mm
001	0.03	.120	3.05
002	0.05	.182	4.62
005	0.13	.197	5.00
.006	0.15	.200	5.08
.008	0.20	.210	5.33
.010	0.25	218	5.54
015	0.38	.275	6.98
.020	0.51	.300	7.62
025	0.64	.344	8.74
.042	1.07	.375	9 52
.050	1.27	.407	10.34
.054	1.37	.462	11 73
.085	2.16	.485	12.32
.088	2.24	.662	16.81
.100	2.54	.882	22.40

NOTES.
1. Dimensions are in inches
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerance is ±.005 (0.13 mm) on decimals and ±2° on angles
4. These connectors mate with connectors specified in MIL-C-55302/177 and /184.
5. Location indicators embossed on surface
6. Numbers indicating end cavities, letters indicating row nearest to polarizing feature, and markings every ten positions stamped on this surface.
7. Mounting ears and hardware are purchased separately. See MIL-C-55302/182 for mounting hardware.
8. Chamfers shall be .015 (0.38 mm) minimum at a 45° angle ±5°.

FIGURE 1. Connectors, pin assemblies, right angle, 100 (2.54 mm) spacing, 75 through 225 contacts - Continued

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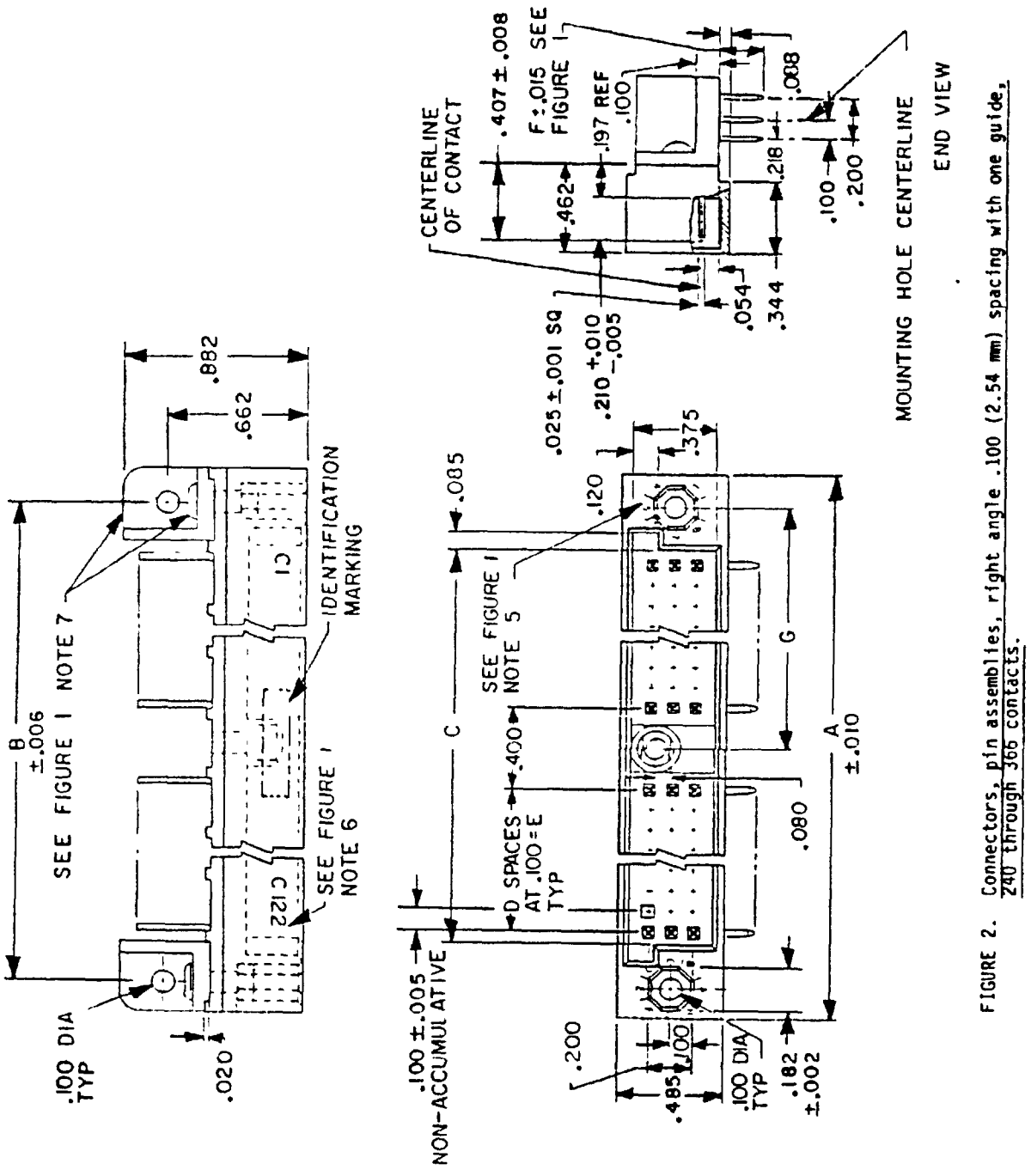
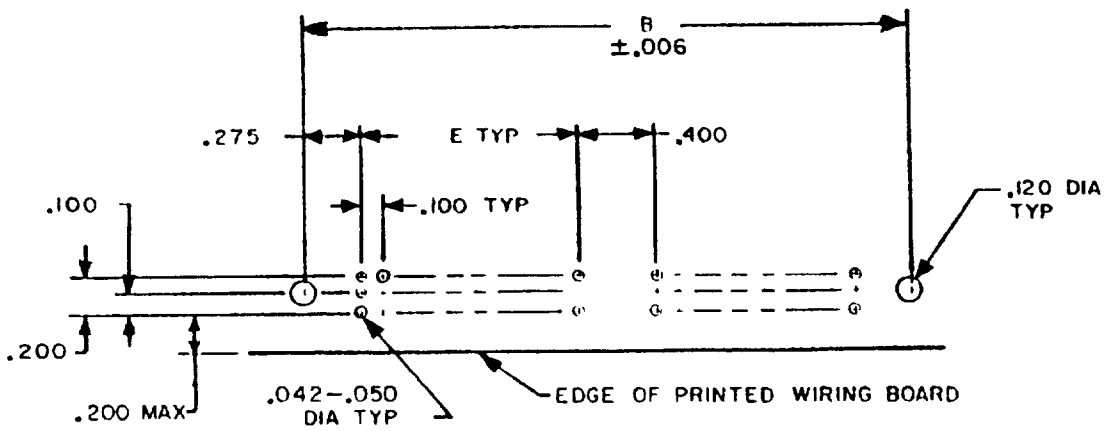


FIGURE 2. Connectors, pin assemblies, right angle .100 (2.54 mm) spacing with one guide, 240 through 366 contacts.

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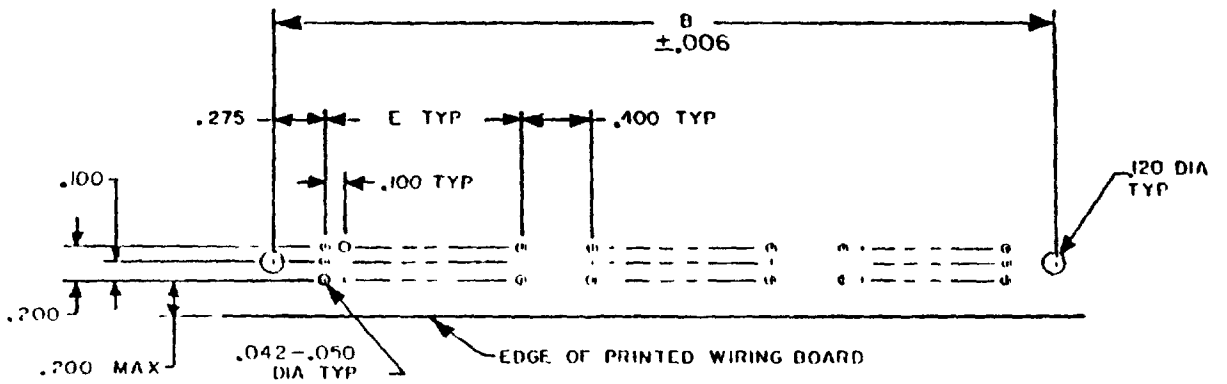
Recommended circuit layout

Inches	mm	Inches	mm
.001	0.03	.100	2.54
.002	0.05	.120	3.05
.005	0 13	.182	4.62
.006	0 15	.197	5.00
.008	0.20	.200	5.08
.010	0.25	.210	5.33
.015	0 38	.218	5.54
.020	0.51	.275	6.98
.025	0.64	.344	8.74
.042	1.07	.375	9.52
.050	1.27	.400	10.16
.054	1.37	.407	10.34
.080	2.03	.462	11.73
.085	2.16	.662	16.81
.088	2.24	.882	22.40

NOTES
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerance is ±.005 (0 13 mm) on decimals and ±2° on angles
4. With one guide, otherwise same notes as figure 1
5. Guide pins are purchased separately. See MIL-C-55302/182.

FIGURE 2. Connectors, pin assemblies, right angle .100 (2.54 mm) spacing with one guide, 240 through 366 contacts - Continued.

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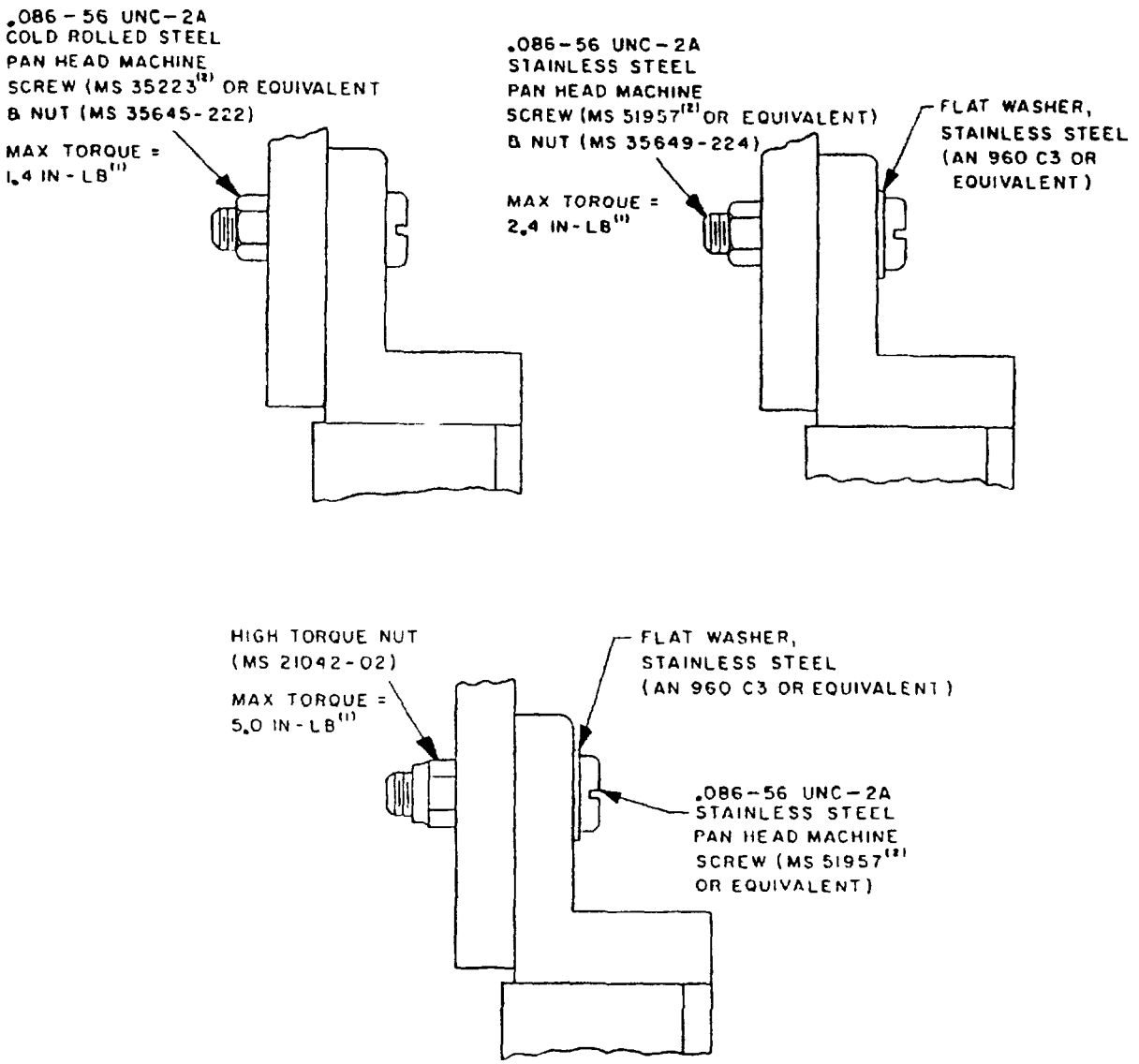


Inches	mm
.006	0.15
.042	1.07
.050	1.27
.100	2.54
.120	3.05
.200	5.08
.275	6.98
.400	10.16

NOTES
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerance is ±.005 (0.13 mm) on decimals and ±2° on angles.
4. With two guides, otherwise same notes as figure 1.
5. Guide pins are purchased separately. See MIL-C-55302/182.

FIGURE 3 Connectors, pin assemblies, right angle .100 (2 54 mm) spacing with two guides, 387 through 405 contacts.

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NOTES
1. Torque wrench must be used when fastening connector to printed circuit board.
2. Length to be determined by user

FIGURE 4. Mounting recommendations.

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TABLE I. Dash number, number of contacts, contact type, and dimensions. 1/

Dash no.	No. of con-tacts	Fig. no.	Contact type	A	B	C	D	E	G	H	Contact ident. no.
01 04	75	1	I IV	3.250 (82.55)	2.950 (74.93)	2.550 (64.77)	24	2.400 (60.96)			1-25
09 12	96	1	I IV	3.950 (100.33)	3.650 (92.71)	3.250 (82.55)	31	3.100 (76.74)			1-32
17 20	120	1	I IV	4.750 (120.65)	4.450 (113.03)	4.050 (102.87)	39	3.900 (99.06)			1-40
25 28	150	1	I IV	5.750 (146.05)	5.450 (138.43)	5.050 (128.27)	49	4.900 (124.46)			1-50
29 32	165	1	I IV	6.250 (158.75)	5.950 (151.13)	5.550 (140.97)	54	5.400 (137.16)			1-55
33 36	180	1	I IV	6.750 (171.45)	6.450 (163.83)	6.050 (153.67)	59	5.900 (149.86)			1-60
41 44	201	1	I IV	7.450 (189.23)	7.150 (101.61)	6.750 (171.45)	66	6.600			1-67
45 48	210	1	I IV	7.750 (196.85)	7.450 (189.23)	7.050 (179.07)	69	6.900 (175.26)			1-70
53 56	240	2	I IV	9.050 (229.87)	8.750 (222.25)	8.350 (212.09)	39	3.900 (99.06)	4.375 (111.13)		1-80
61 64	276	2	I IV	10.250 (260.35)	9.950 (252.73)	9.550 (242.57)	45	4.500 (114.30)	4.975 (126.37)		1-92
73 76	330	2	I IV	12.050 (306.07)	11.750 (298.45)	11.350 (288.29)	54	5.400 (137.16)	5.875 (149.23)		1-110
81 84	366	2	I IV	13.250 (336.55)	12.950 (328.93)	12.550 (318.77)	60	6.000 (152.40)	6.475 (164.47)		1-222
88 92	405	3 2/	I IV	14.850 (377.19)	14.550 (369.57)	14.150 (359.41)	44	4.400 (111.76)	4.875 (123.83)	4.800 (121.92)	1-135

1/ Deleted dash numbers in the series 01 through 92 are canceled with no replacement.
2/ For more information, see figure 3 on MIL-C-55302/177.

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TABLE II. Contact plating requirements. 1/

Type	Finish requirement
A	Overall contact finish: Shall be gold in accordance with MIL-C-55302.
B	Localized contact finish. Contact engagement area shall have a minimum plating in accordance with MIL-C-55302. Solder post area (contact type I, .195 minimum length; type IV, .135 minimum length) shall be tin-lead over nickel in accordance with MIL-C-55302.

1/ Type C finish is deleted and superseded by type A or B.

REQUIREMENTS:

Design and construction:

Dimensions and configuration. See figures 1, 2, 3, 4, and table I.

Material: In accordance with MIL-C-55302.

Plating. See table II

Current rating: 3.0 amperes maximum per contact, 2.25 amperes continuous per contact at room ambient with no more than two adjacent contacts carrying this current.

Keying (see MIL-C-55302/31):

Two keys, Part or Identifying Number (PIN) MS55302/31-04, and two .086-56 UNC-2A mounting screws are recommended. Keys and screws shall be ordered separately

Mating and unmating. The maximum mating force in pounds shall be the number of contacts multiplied by .125, the withdrawal force in pounds shall be a minimum of .025 times the number of contacts and shall not exceed the measured insertion force.

Contact resistance No individual contact pair shall have a resistance exceeding .015 ohm initial and .020 ohm after testing

Contact retention: 3 pounds, minimum.

Dielectric withstanding voltage:

Sea level. 900 volts rms.

70,000 feet: 200 volts rms

2 milliamperes maximum leakage current.

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Part or Identifying Number (PIN) M55302/178 (and dash number from table I and type from table II).

Example M55302/178-01A.

Revision letters are not used to denote changes due to the extensiveness of the changes.

CONCLUDING MATERIAL

Custodians.
Army - CR
Navy - EC
Air Force - 17

Review activities
Army - AR, MI
Navy - AS, SH
Air Force - 11, 85, 99
DLA - ES

User activities·
Army - AT, AV, ME
Navy - MC
Air Force - 19

Preparing activity.
Air Force - 17

Agent·
DLA - ES

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